

The author doesn't explicitly state a research question but based on his datasets and wrangling, sequence of analyses and confidence intervals generated, he seems to be answering: why were more goals scored in the 2023–24 Premier League season, and was the increase associated with longer added time and more goals being scored after 90 minutes?

I perform the same data wrangling and analyses as well as extending the investigation to see if any other match characteristics can be associated with the change, and how might a changing scoring environment affect bookmaker goal-market probabilities and predictions for future seasons?

This will be especially relevant to markets such as over/under 2.5 goals and both teams to score.

```
library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.5.3
## Warning: package 'readr' was built under R version 4.5.3
## Warning: package 'forcats' was built under R version 4.5.3
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.2.0
## ✓ forcats   1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.1      ✓ tibble     3.3.0
## ✓ lubridate 1.9.4      ✓ tidyr      1.3.1
## ✓ purrr     1.2.0
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(BSDA)

## Warning: package 'BSDA' was built under R version 4.5.3
## Loading required package: lattice
##
## Attaching package: 'BSDA'
##
## The following object is masked from 'package:datasets':
##
##      Orange
```

```

library(dplyr)
knitr::opts_chunk$set(
  warning = FALSE,
  message = FALSE
)

MatchData_2021_22 <- read_csv("~/Data Science Toolbox/21-22.csv", show_col_types = FALSE)
MatchData_2022_23 <- read_csv("~/Data Science Toolbox/22-23.csv", show_col_types = FALSE)
MatchData_2023_24 <- read_csv("~/Data Science Toolbox/23-24.csv", show_col_types = FALSE)

MatchEvents_2021 <- read_csv("~/Data Science Toolbox/2021-22_match id home away and event type.csv") %>%
  mutate(Season = "2021-22")

MatchEvents_2022 <- read_csv("~/Data Science Toolbox/2022-23_match id home away and event type.csv") %>%
  mutate(Season = "2022-23")

MatchEvents_2023 <- read_csv("~/Data Science Toolbox/2023-24_match id home away and event type.csv") %>%
  mutate(Season = "2023-24")

# Add season column and calculate total goals per match

MatchData_2021_22 <- MatchData_2021_22 %>%
  mutate(
    season = "2021-22",
    total_goals = home_team_goals + away_team_goals
  )

MatchData_2022_23 <- MatchData_2022_23 %>%
  mutate(
    season = "2022-23",
    total_goals = home_team_goals + away_team_goals
  )

MatchData_2023_24 <- MatchData_2023_24 %>%
  mutate(
    season = "2023-24",
    total_goals = home_team_goals + away_team_goals
  )

# Combine match data for all three seasons
combined_matchdata <- bind_rows(
  MatchData_2021_22,
  MatchData_2022_23,
  MatchData_2023_24
)

```

```

)

# Combine match-event data for all three seasons
combined_matchevents <- bind_rows(
  MatchEvents_2021,
  MatchEvents_2022,
  MatchEvents_2023
)

dim(combined_matchdata)
## [1] 1140    13

dim(combined_matchevents)
## [1] 16015     5

table(combined_matchdata$season)
##
## 2021-22 2022-23 2023-24
##    380    380    380

table(combined_matchevents$Season)
##
## 2021-22 2022-23 2023-24
##   4562   5488   5965

# Check variable types in match data

class(combined_matchdata$`45+`)
## [1] "numeric"

class(combined_matchdata$`90+`)
## [1] "numeric"

class(combined_matchdata$Match_Id)
## [1] "numeric"

class(combined_matchdata$match_date)
## [1] "character"

class(combined_matchdata$home_team_goals)
## [1] "numeric"

class(combined_matchdata$away_team_goals)
## [1] "numeric"

```

```

class(combined_matchdata$attendance)
## [1] "numeric"
combined_matchdata$match_date <- as.Date(combined_matchdata$match_date)
#combined_matchdata <- combined_matchdata %>%
# mutate(attendance = parse_number(attendance))

```

Here I found that in the original authors dataset there were character strings in the attendance column, he ran code that removed it but I removed it manually from the excel sheet, although this column wasn't used in any analysis anyway.

```

class(combined_matchdata$match_date)
## [1] "Date"
class(combined_matchdata$attendance)
## [1] "numeric"
sum(is.na(combined_matchdata$attendance))
## [1] 78
sum(is.na(MatchData_2021_22$attendance))
## [1] 23
sum(is.na(MatchData_2022_23$attendance))
## [1] 30
sum(is.na(MatchData_2023_24$attendance))
## [1] 25
class(combined_matchevents$Match_Id)
## [1] "numeric"
class(combined_matchevents$Home_Away)
## [1] "character"
class(combined_matchevents$Event_Minute)
## [1] "numeric"
class(combined_matchevents$Event_Type)
## [1] "character"
unique(combined_matchevents$Event_Type)

```

```

## [1] "Goal" "Substitution"
## [3] "Yellow Card" "label.penalty.scored"
## [5] "Own Goal" "Red Card"
## [7] "Second Yellow Card (Red Card)" "Substitution Off"

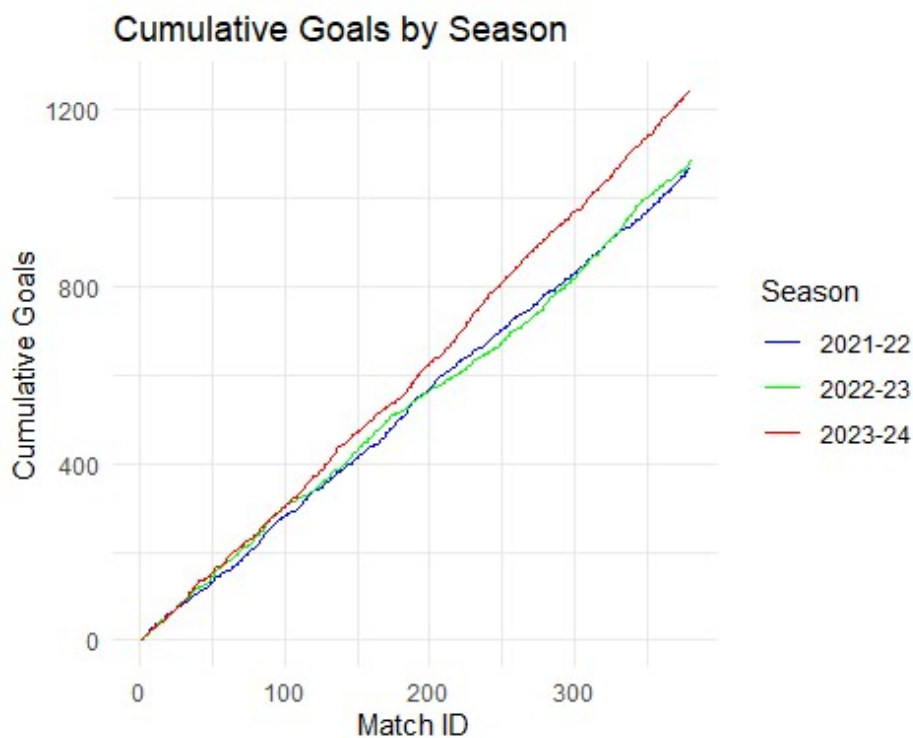
combined_matchevents <- combined_matchevents %>%
  mutate(
    Event_Type = if_else(
      Event_Type == "Second Yellow Card (Red Card)",
      "Red Card",
      Event_Type
    )
  )
unique(combined_matchevents$Event_Type)

## [1] "Goal" "Substitution" "Yellow Card"
## [4] "label.penalty.scored" "Own Goal" "Red Card"
## [7] "Substitution Off"

combined_matchdata <- combined_matchdata %>%
  group_by(season) %>%
  arrange(Match_Id) %>%
  mutate(cumulative_goals = cumsum(total_goals)) %>%
  ungroup()

ggplot(
  combined_matchdata,
  aes(x = Match_Id, y = cumulative_goals, colour = season)
) +
  geom_line() +
  scale_colour_manual(
    values = c(
      "2021-22" = "blue",
      "2022-23" = "green",
      "2023-24" = "red"
    )
  ) +
  labs(
    title = "Cumulative Goals by Season",
    x = "Match ID",
    y = "Cumulative Goals",
    colour = "Season"
  ) +
  theme_minimal()

```



```
total_goals_per_season <- combined_matchevents %>%
  filter(Event_Type %in% c(
    "Goal",
    "Own Goal",
    "label.penalty.scored"
  )) %>%
  group_by(Season) %>%
  summarise(Total_Goals = n())
```

```
total_goals_per_season
```

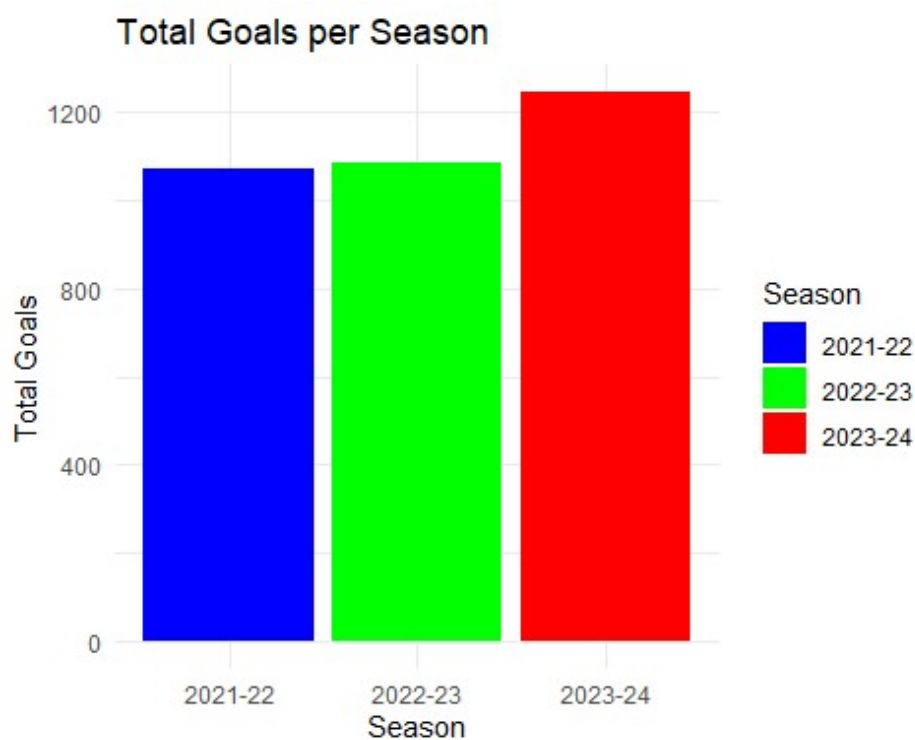
```
## # A tibble: 3 × 2
##   Season Total_Goals
##   <chr>      <int>
## 1 2021-22      1071
## 2 2022-23      1084
## 3 2023-24      1246
```

```
combined_matchdata %>%
  group_by(season) %>%
  summarise(
    Total_Goals = sum(total_goals)
  )
```

```
## # A tibble: 3 × 2
##   season Total_Goals
##   <chr>      <dbl>
```

```
## 1 2021-22      1071
## 2 2022-23      1084
## 3 2023-24      1246

ggplot(total_goals_per_season,
       aes(x = Season, y = Total_Goals, fill = Season)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Total Goals per Season",
    x = "Season",
    y = "Total Goals"
  ) +
  theme_minimal()
```



```
goal_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    mean_goals = mean(total_goals),
    sd_goals = sd(total_goals),
    median_goals = median(total_goals)
  )
```

```
goal_summary
```

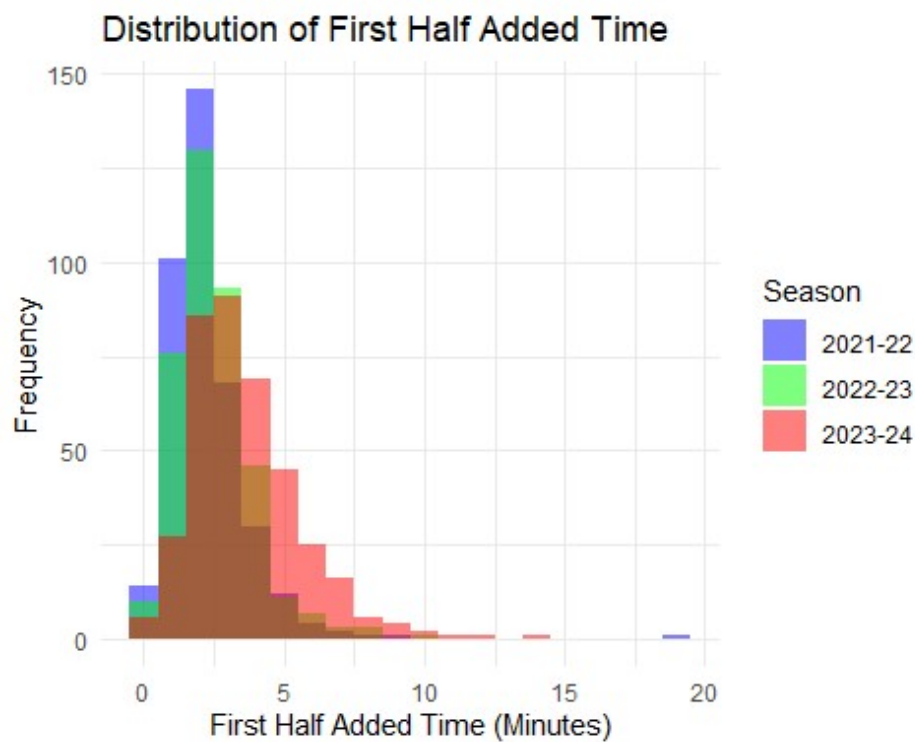
```
## # A tibble: 3 × 4
##   season mean_goals sd_goals median_goals
##   <chr>      <dbl>    <dbl>         <dbl>
## 1 2021-22      2.82      1.63           3
## 2 2022-23      2.85      1.79           3
## 3 2023-24      3.28      1.66           3
```

```
added_time_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    mean_45_added = mean(`45+`),
    sd_45_added = sd(`45+`),
    mean_90_added = mean(`90+`),
    sd_90_added = sd(`90+`)
  )
```

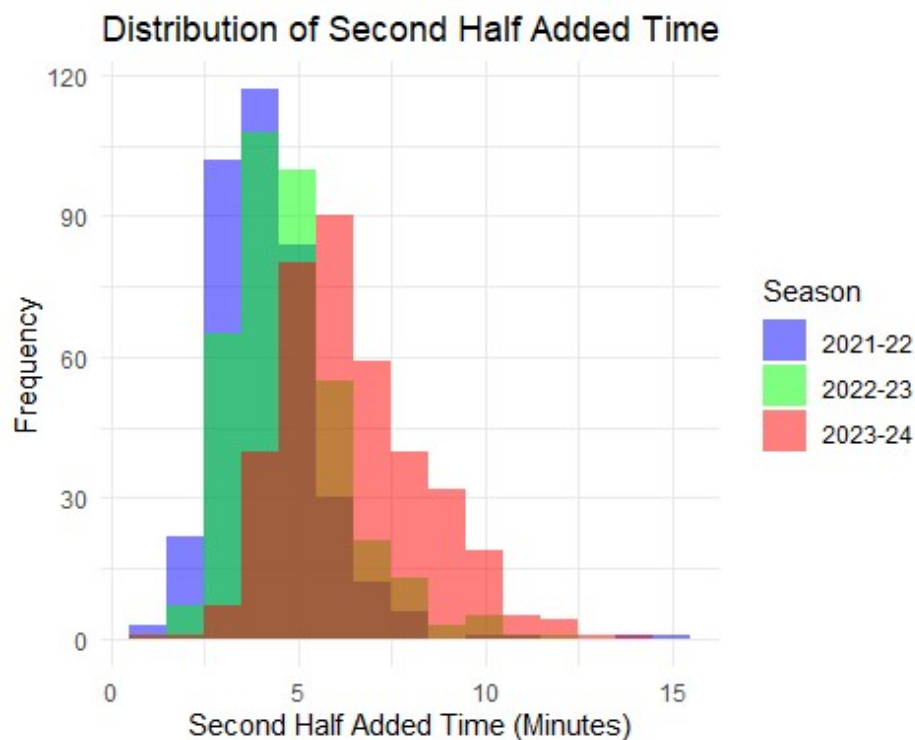
```
added_time_summary
```

```
## # A tibble: 3 × 5
##   season mean_45_added sd_45_added mean_90_added sd_90_added
##   <chr>      <dbl>    <dbl>         <dbl>         <dbl>
## 1 2021-22      2.24      1.56           4.22           1.55
## 2 2022-23      2.50      1.41           4.80           1.58
## 3 2023-24      3.62      1.98           6.49           1.95
```

```
ggplot(combined_matchdata,
  aes(x = `45+`, fill = season)) +
  geom_histogram(
    position = "identity",
    alpha = 0.5,
    binwidth = 1
  ) +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Distribution of First Half Added Time",
    x = "First Half Added Time (Minutes)",
    y = "Frequency",
    fill = "Season"
  ) +
  theme_minimal()
```



```
ggplot(combined_matchdata,
       aes(x = `90+`, fill = season)) +
  geom_histogram(
    position = "identity",
    alpha = 0.5,
    binwidth = 1
  ) +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Distribution of Second Half Added Time",
    x = "Second Half Added Time (Minutes)",
    y = "Frequency",
    fill = "Season"
  ) +
  theme_minimal()
```



How many goals were recorded after minute 90 in each season?

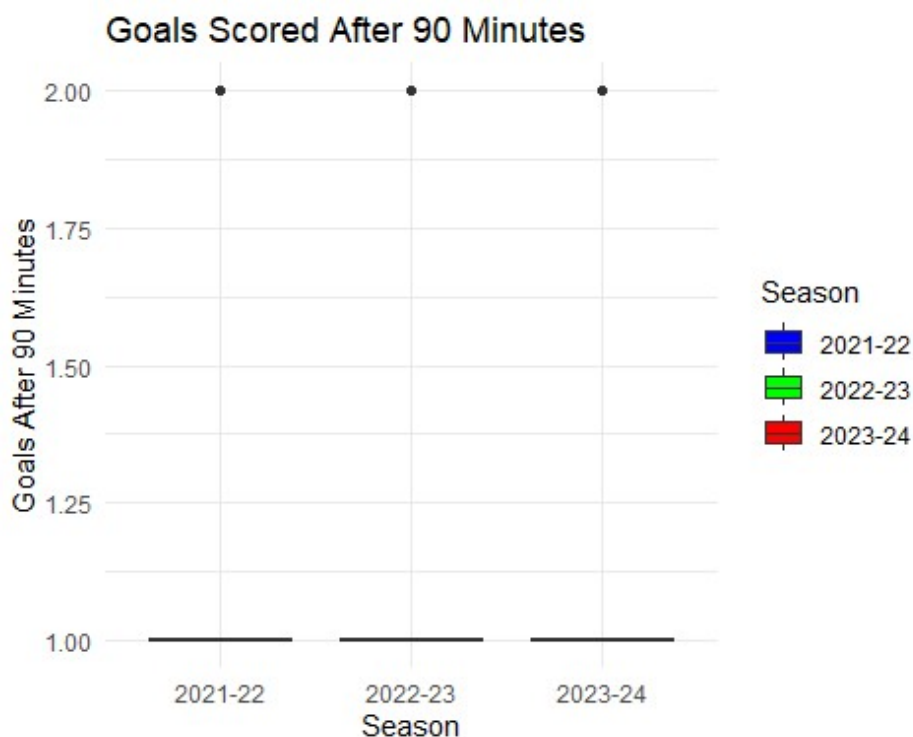
```
goals_after_90_per_match <- combined_matchevents %>%
  filter(
    Event_Type %in% c(
      "Goal",
      "Own Goal",
      "label.penalty.scored"
    ),
    Event_Minute > 90
  ) %>%
  group_by(Season, Match_Id) %>%
  summarise(goals_after_90 = n())
```

#goals_after_90

```
head(goals_after_90_per_match)
```

```
## # A tibble: 6 × 3
## # Groups:   Season [1]
##   Season Match_Id goals_after_90
##   <chr>      <dbl>      <int>
## 1 2021-22      10          1
## 2 2021-22      26          2
## 3 2021-22      33          1
## 4 2021-22      34          1
## 5 2021-22      36          1
## 6 2021-22      38          1
```

```
ggplot(
  goals_after_90_per_match,
  aes(x = Season, y = goals_after_90, fill = Season)
) +
  geom_boxplot() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Goals Scored After 90 Minutes",
    x = "Season",
    y = "Goals After 90 Minutes"
  ) +
  theme_minimal()
```



This boxplot is not particularly informative because most matches containing a post-90 goal record only one such goal. A more useful comparison is therefore the percentage of matches in each season containing at least one goal after the 90th minute.

```
late_goal_percentage <- combined_matchevents %>%
  filter(
    Event_Type %in% c(
      "Goal",
      "Own Goal",
      "label.penalty.scored"
    )
  )
```

```

    ),
    Event_Minute > 90
  ) %>%
  group_by(Season) %>%
  summarise(
    matches_with_late_goal = n_distinct(Match_Id),
    percentage = 100 * matches_with_late_goal / 380
  )

```

late_goal_percentage

```

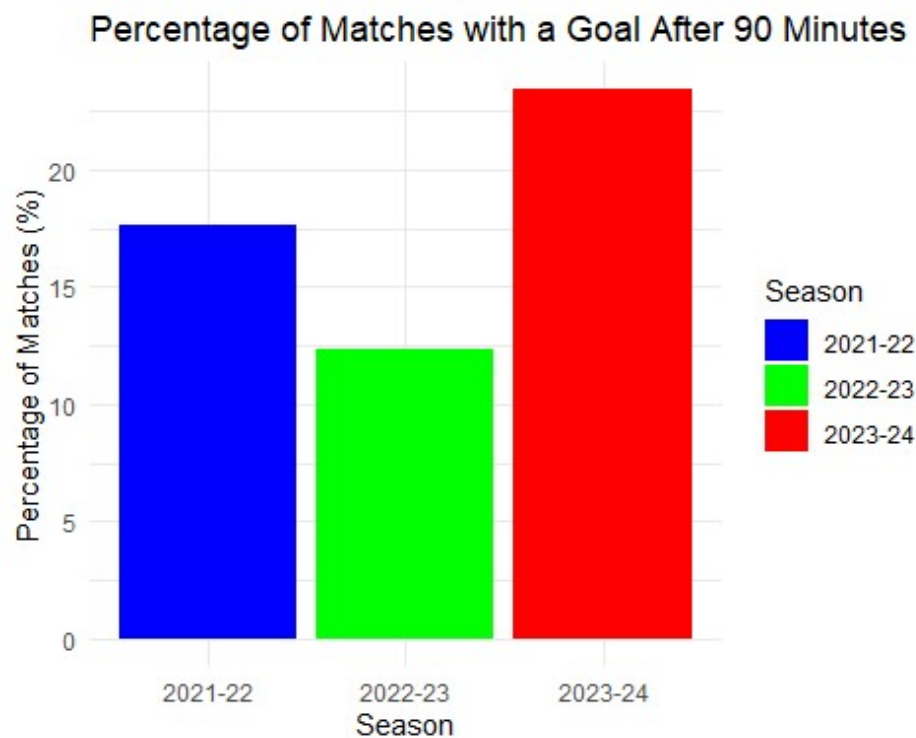
## # A tibble: 3 × 3
##   Season matches_with_late_goal percentage
##   <chr>          <int>          <dbl>
## 1 2021-22             67           17.6
## 2 2022-23             47           12.4
## 3 2023-24             89           23.4

```

```

ggplot(
  late_goal_percentage,
  aes(x = Season, y = percentage, fill = Season)
) +
  geom_col() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Percentage of Matches with a Goal After 90 Minutes",
    x = "Season",
    y = "Percentage of Matches (%)"
  ) +
  theme_minimal()

```



Among matches where a post-90 goal occurred, how many post-90 goals occurred on average?

```
late_goal_summary <- goals_after_90_per_match %>%
  group_by(Season) %>%
  summarise(
    mean_goals_after_90 = mean(goals_after_90),
    sd_goals_after_90 = sd(goals_after_90)
  )

late_goal_summary

## # A tibble: 3 × 3
##   Season mean_goals_after_90 sd_goals_after_90
##   <chr>         <dbl>         <dbl>
## 1 2021-22         1.07         0.265
## 2 2022-23         1.04         0.204
## 3 2023-24         1.16         0.366

goals_after_90_all_matches <- combined_matchdata %>%
  select(season, Match_Id) %>%
  left_join(
    goals_after_90_per_match,
    by = c("season" = "Season", "Match_Id" = "Match_Id")
  ) %>%
  mutate(
    goals_after_90 = replace_na(goals_after_90, 0)
  )
```

```
late_goal_summary_all <- goals_after_90_all_matches %>%
  group_by(season) %>%
  summarise(
    mean_goals_after_90 = mean(goals_after_90),
    sd_goals_after_90 = sd(goals_after_90)
  )
```

```
late_goal_summary_all
```

```
## # A tibble: 3 × 3
##   season mean_goals_after_90 sd_goals_after_90
##   <chr>          <dbl>          <dbl>
## 1 2021-22          0.189          0.425
## 2 2022-23          0.129          0.351
## 3 2023-24          0.271          0.522
```

```
late_goal_summary_all <- goals_after_90_all_matches %>%
  group_by(season) %>%
  summarise(
    mean_goals_after_90 = mean(goals_after_90),
    sd_goals_after_90 = sd(goals_after_90)
  )
```

```
late_goal_summary_all
```

```
## # A tibble: 3 × 3
##   season mean_goals_after_90 sd_goals_after_90
##   <chr>          <dbl>          <dbl>
## 1 2021-22          0.189          0.425
## 2 2022-23          0.129          0.351
## 3 2023-24          0.271          0.522
```

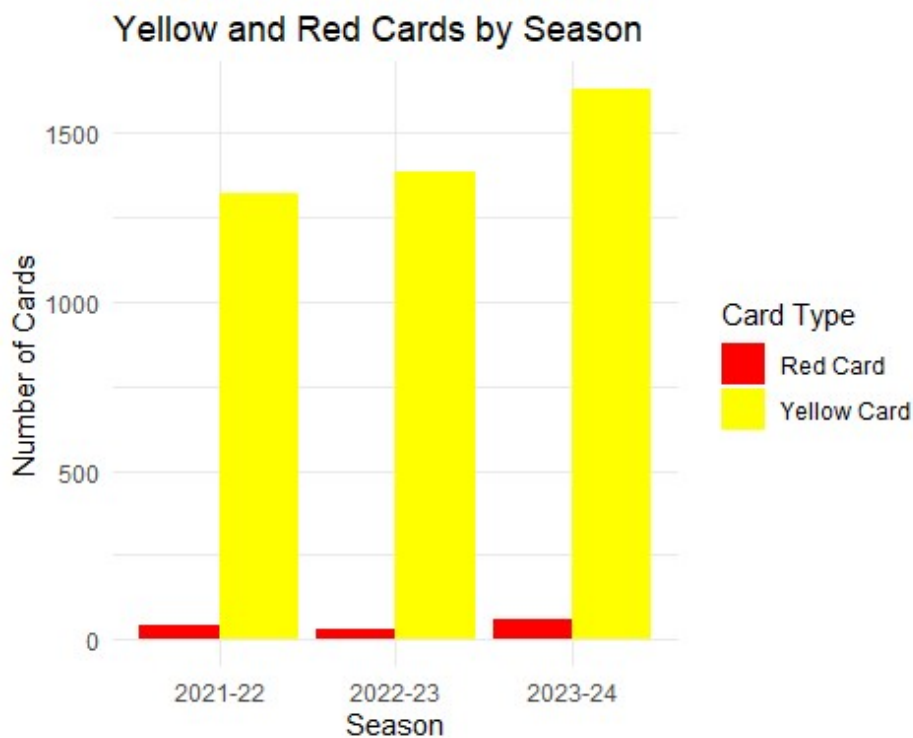
```
cards_per_season <- combined_matchevents %>%
  filter(Event_Type %in% c(
    "Yellow Card",
    "Red Card"
  )) %>%
  group_by(Season, Event_Type) %>%
  summarise(
    Total_Cards = n()
  )
```

```
cards_per_season
```

```
## # A tibble: 6 × 3
## # Groups:   Season [3]
##   Season Event_Type Total_Cards
##   <chr>   <chr>         <int>
## 1 2021-22 Red Card           43
## 2 2021-22 Yellow Card       1320
## 3 2022-23 Red Card           30
```

```
## 4 2022-23 Yellow Card      1387
## 5 2023-24 Red Card         58
## 6 2023-24 Yellow Card     1631
```

```
ggplot(
  cards_per_season,
  aes(x = Season, y = Total_Cards, fill = Event_Type)
) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c(
    "Yellow Card" = "yellow",
    "Red Card" = "red"
  )) +
  labs(
    title = "Yellow and Red Cards by Season",
    x = "Season",
    y = "Number of Cards",
    fill = "Card Type"
  ) +
  theme_minimal()
```



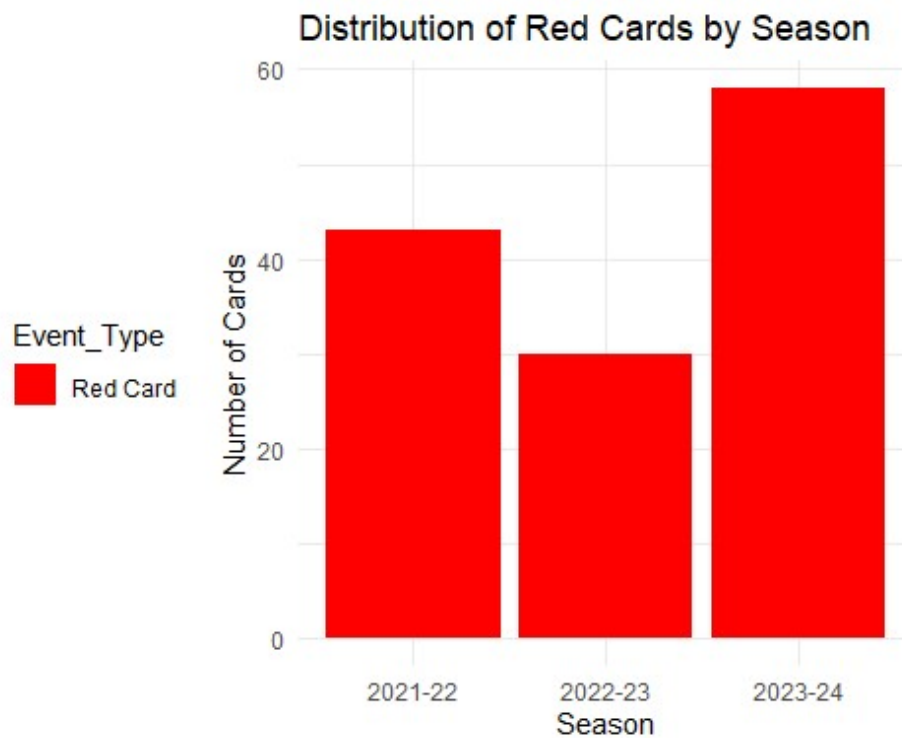
```
Red_Cards_per_season <- combined_matchevents %>%
  filter(Event_Type == "Red Card")

ggplot(Red_Cards_per_season, aes(x = Season, fill = Event_Type)) +
  geom_bar() +
  labs(
```

```

    title = "Distribution of Red Cards by Season",
    x = "Season",
    y = "Number of Cards"
  ) +
  theme_minimal() +
  scale_fill_manual(values = c("Red Card" = "red")) +
  theme(legend.position = "left")

```

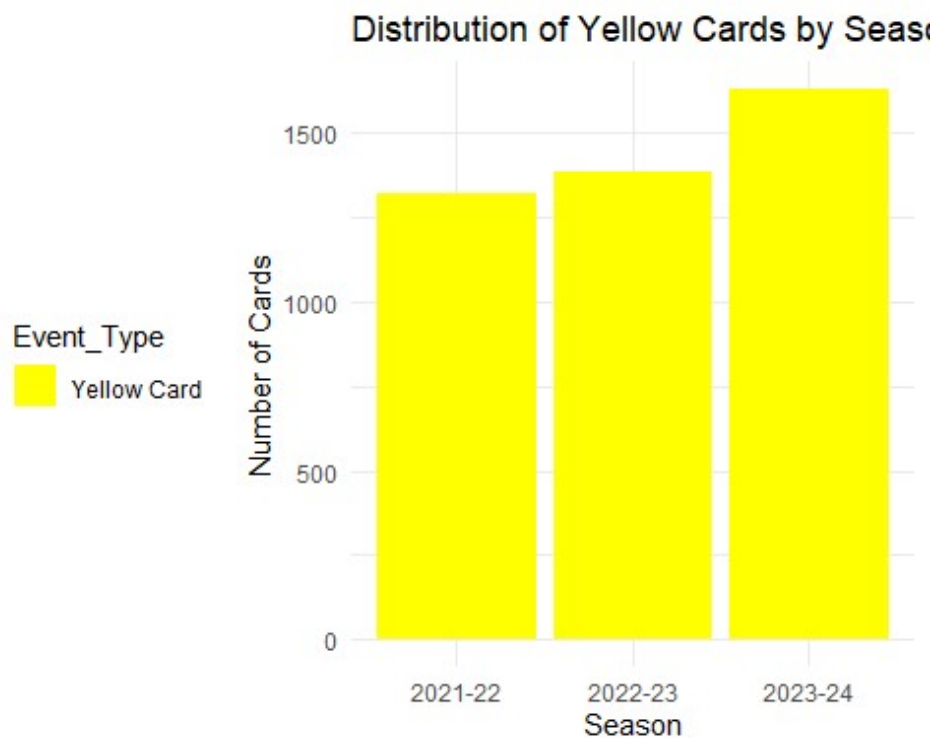


```

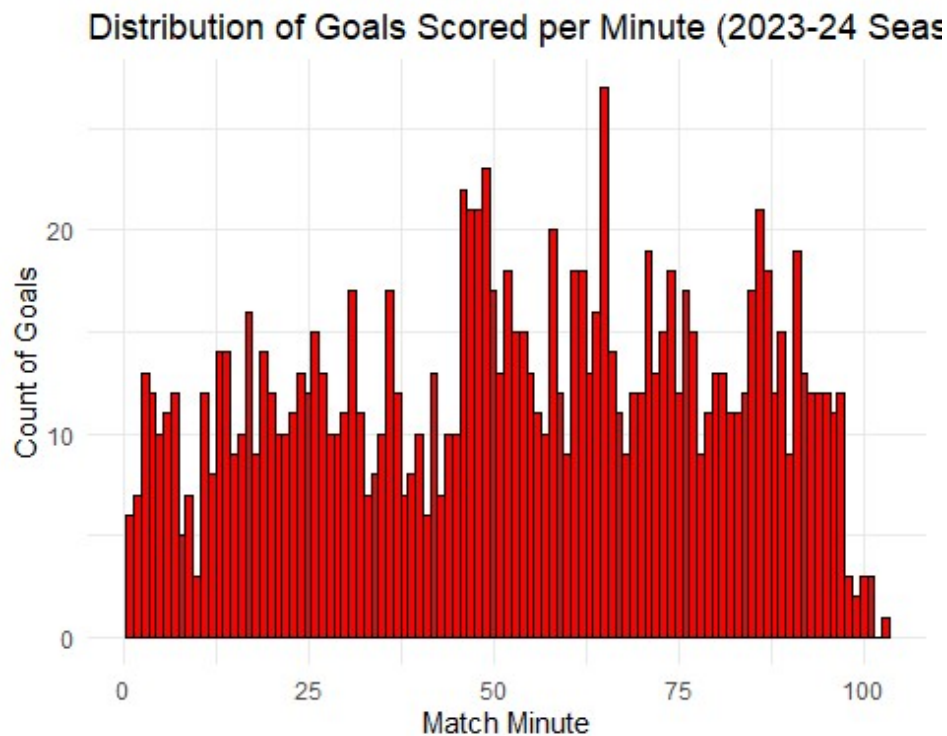
Yellow_Cards_per_season <- combined_matchevents %>%
  filter(Event_Type == "Yellow Card")

ggplot(Yellow_Cards_per_season, aes(x = Season, fill = Event_Type)) +
  geom_bar() +
  labs(
    title = "Distribution of Yellow Cards by Season",
    x = "Season",
    y = "Number of Cards"
  ) +
  theme_minimal() +
  scale_fill_manual(values = c("Yellow Card" = "yellow")) +
  theme(legend.position = "left")

```



```
Goals_2023_24 <- combined_matchevents %>%
  filter(
    Season == "2023-24" &
    Event_Type %in% c(
      "Goal",
      "Own Goal",
      "label.penalty.scored"
    )
  )
ggplot(Goals_2023_24, aes(x = Event_Minute)) +
  geom_histogram(
    binwidth = 1,
    fill = "red",
    color = "black"
  ) +
  labs(
    title = "Distribution of Goals Scored per Minute (2023-24 Season)",
    x = "Match Minute",
    y = "Count of Goals"
  ) +
  theme_minimal()
```



```
Goals_after_90_2023_24 <- combined_matchevents %>%
  filter(
    Season == "2023-24" &
    Event_Minute > 90 &
    Event_Type %in% c(
      "Goal",
      "Own Goal",
      "label.penalty.scored"
    )
  )

ggplot(Goals_after_90_2023_24, aes(x = Event_Minute)) +
  geom_histogram(
    binwidth = 1,
    fill = "red",
    color = "black"
  ) +
  labs(
    title = "Distribution of Goals Scored After 90 Minutes (2023-24 Season)",
    x = "Extra Time Minute",
    y = "Count of Goals"
  ) +
  theme_minimal() +
  scale_x_continuous(
    breaks = seq(
      91,
      max(Goals_after_90_2023_24$Event_Minute),
```

```

    by = 5
  )
)

```

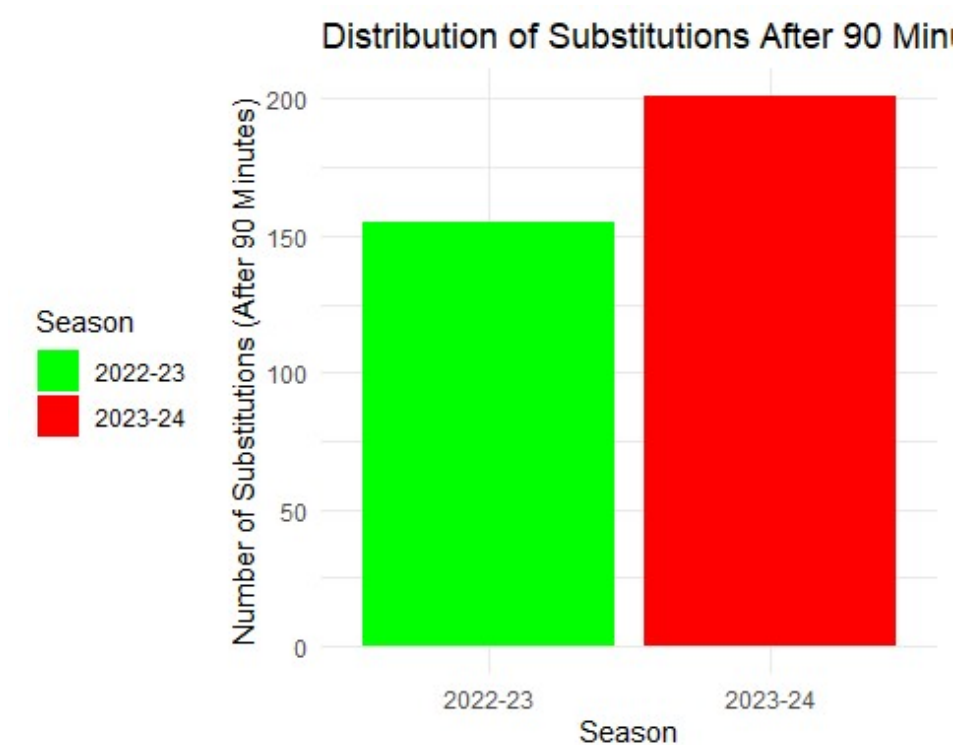


```

Substitution_after_90 <- combined_matchevents %>%
  filter(
    Event_Type == "Substitution" &
    Event_Minute > 90 &
    Season %in% c("2022-23", "2023-24")
  )

ggplot(Substitution_after_90, aes(x = Season, fill = Season)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Distribution of Substitutions After 90 Minutes by Season",
    x = "Season",
    y = "Number of Substitutions (After 90 Minutes)"
  ) +
  theme_minimal() +
  scale_fill_manual(values = c(
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  theme(legend.position = "left")

```



```
combined_matchdata <- combined_matchdata %>%
  mutate(
    total_playing_time = 90 + `45+` + `90+`
  )

names(combined_matchdata)

## [1] "Match_Id"          "match_date"
## [3] "attendance"        "home_team"
## [5] "away_team"         "home_team_goals"
## [7] "away_team_goals"   "home_team_goals_half_time"
## [9] "away_team_goals_half_time" "45+"
## [11] "90+"              "season"
## [13] "total_goals"       "cumulative_goals"
## [15] "total_playing_time"

playing_time_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    mean_playing_time = mean(total_playing_time),
    sd_playing_time = sd(total_playing_time)
  )

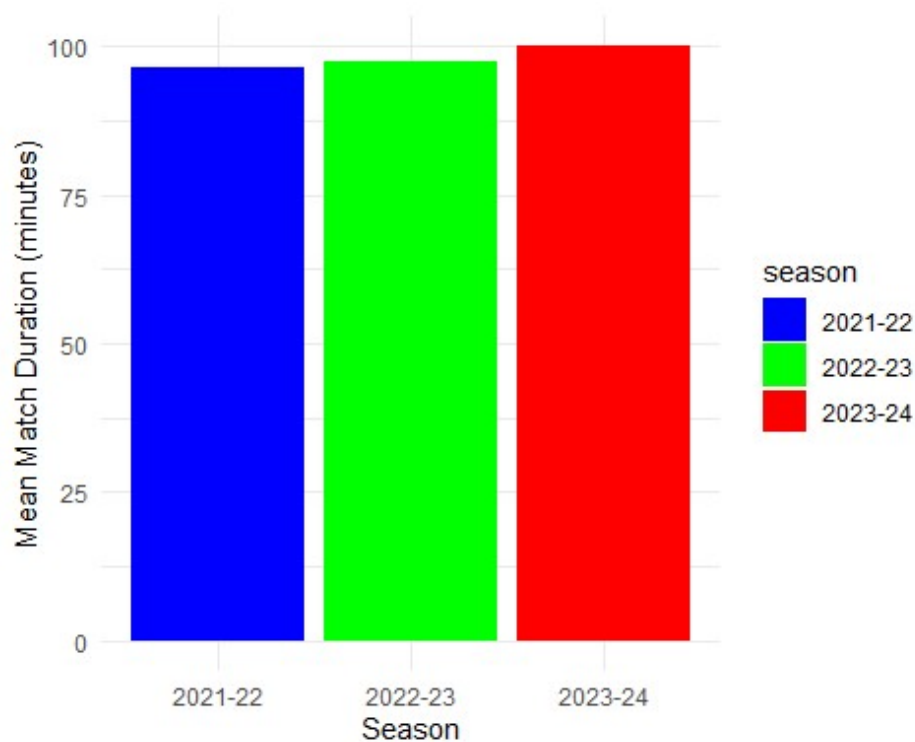
playing_time_summary

## # A tibble: 3 × 3
##   season mean_playing_time sd_playing_time
```

```
##      <chr>          <dbl>      <dbl>
## 1 2021-22          96.5        2.31
## 2 2022-23          97.3        2.31
## 3 2023-24         100.         2.66

ggplot(
  playing_time_summary,
  aes(x = season, y = mean_playing_time, fill = season)
) +
  geom_col() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    x = "Season",
    y = "Mean Match Duration (minutes)"
  ) +
  theme_minimal()

```



```
summary_table_count <- combined_matchevents %>%
  group_by(Match_Id, Season) %>%
  summarise(
    goal_count = sum(
      Event_Type %in% c(
        "Goal",

```

```

      "label.penalty.scored",
      "Own Goal"
    )
  ),
  red_card_count = sum(Event_Type == "Red Card"),
  yellow_card_count = sum(Event_Type == "Yellow Card"),
  substitution_count = sum(Event_Type == "Substitution")
)

head(summary_table_count)

## # A tibble: 6 × 6
## # Groups:   Match_Id [2]
##   Match_Id Season goal_count red_card_count yellow_card_count substitution
##   <dbl> <chr>      <int>          <int>          <int>          <int>
## 1      1 2021-...      2            0            0            0
## 2      1 2022-...      2            0            3            0
## 3      1 2023-...      3            1            0            0
## 4      2 2021-...      3            0            3            0
## 5      2 2022-...      2            0            6            0
## 6      2 2023-...      3            0            4            0

summary_table_count_90 <- combined_matchevents %>%
  filter(Event_Minute > 90) %>%
  group_by(Match_Id, Season) %>%
  summarise(
    goal_count = sum(
      Event_Type %in% c(
        "Goal",
        "label.penalty.scored",
        "Own Goal"
      )
    ),
    red_card_count = sum(Event_Type == "Red Card"),
    yellow_card_count = sum(Event_Type == "Yellow Card"),
    substitution_count = sum(Event_Type == "Substitution")
  )

dim(summary_table_count)

## [1] 1140      6

table(summary_table_count_90$Season)

```

```

##
## 2021-22 2022-23 2023-24
##      203      226      276

all_matches <- combined_matchevents %>%
  distinct(Match_Id, Season)

summary_count_90_complete <- combined_matchevents %>%
  filter(Event_Minute > 90) %>%
  group_by(Match_Id, Season) %>%
  summarise(
    goal_count = sum(
      Event_Type %in% c(
        "Goal",
        "label.penalty.scored",
        "Own Goal"
      )
    ),
    red_card_count = sum(Event_Type == "Red Card"),
    yellow_card_count = sum(Event_Type == "Yellow Card"),
    substitution_count = sum(Event_Type == "Substitution"),
    .groups = "drop"
  ) %>%
  right_join(
    all_matches,
    by = c("Match_Id", "Season")
  ) %>%
  replace_na(list(
    goal_count = 0,
    red_card_count = 0,
    yellow_card_count = 0,
    substitution_count = 0
  )) %>%
  arrange(Season, Match_Id)

dim(summary_count_90_complete)

## [1] 1140      6

table(summary_count_90_complete$Season)

##
## 2021-22 2022-23 2023-24
##      380      380      380

summary_by_match <- combined_matchevents %>%
  group_by(Match_Id, Season) %>%
  summarise(
    goal_count = sum(
      Event_Type %in% c(
        "Goal",

```

```

        "label.penalty.scored",
        "Own Goal"
    )
),
card_count = sum(
    Event_Type %in% c(
        "Red Card",
        "Yellow Card"
    )
),
substitution_count = sum(Event_Type == "Substitution"),
.groups = "drop"
)

combined_summary_2122 <- summary_by_match %>%
  filter(Season %in% c("2021-22", "2022-23"))

combined_summary_2223 <- summary_by_match %>%
  filter(Season %in% c("2022-23", "2023-24"))

dim(combined_summary_2122)
## [1] 760    5

dim(combined_summary_2223)
## [1] 760    5

Mean_of_goals_2021_2022 <- mean(combined_summary_2122$goal_count)

Std_of_goals_2021_2022 <- sd(combined_summary_2122$goal_count)

Mean_of_goals_2021_2022
## [1] 2.835526

Std_of_goals_2021_2022
## [1] 1.709634

```

A 95% confidence interval is calculated for the mean number of goals per match across the 2021–22 and 2022–23 seasons.

```

CI_of_21_23 <- t.test(
  combined_summary_2122$goal_count,
  conf.level = 0.95
)

CI_of_21_23
##
## One Sample t-test

```

```
##
## data: combined_summary_2122$goal_count
## t = 45.723, df = 759, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  2.713785 2.957267
## sample estimates:
## mean of x
##  2.835526

Mean_of_goals_2023_24 <- mean(
  summary_by_match$goal_count[
    summary_by_match$Season == "2023-24"
  ]
)

Mean_of_goals_2023_24

## [1] 3.278947
```

The 2023–24 mean of 3.28 goals per match lies above this confidence interval, indicating that the season had a substantially higher average scoring rate than the previous two seasons.

```
late_goals_21_23 <- summary_count_90_complete %>%
  filter(Season %in% c("2021-22", "2022-23"))

mean(late_goals_21_23$goal_count)

## [1] 0.1592105

sd(late_goals_21_23$goal_count)

## [1] 0.3904923

late_goals_23_24 <- summary_count_90_complete %>%
  filter(Season == "2023-24")

mean(late_goals_23_24$goal_count)

## [1] 0.2710526

sd(late_goals_23_24$goal_count)

## [1] 0.5215199

X_value_goals_after_90_23 <- late_goals_23_24$goal_count
Y_value_goals_after_90_23 <- late_goals_21_23$goal_count

z.test(
  x = X_value_goals_after_90_23,
  y = Y_value_goals_after_90_23,
```

```

mu = 0,
sigma.x = 0.5215199,
sigma.y = 0.3904923,
alternative = "greater"
)

##
## Two-sample z-Test
##
## data: X_value_goals_after_90_23 and Y_value_goals_after_90_23
## z = 3.6946, p-value = 0.0001101
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.06204942 NA
## sample estimates:
## mean of x mean of y
## 0.2710526 0.1592105

```

The original analysis uses a two-sample z-test with standard deviations estimated from the sample data. Since the population variances are unknown, Welch's two-sample t-test is more appropriate.

```

t.test(
  X_value_goals_after_90_23,
  Y_value_goals_after_90_23,
  alternative = "greater"
)

##
## Welch Two Sample t-test
##
## data: X_value_goals_after_90_23 and Y_value_goals_after_90_23
## t = 3.6946, df = 597.81, p-value = 0.0001202
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.06197214 Inf
## sample estimates:
## mean of x mean of y
## 0.2710526 0.1592105

```

The conclusion is unchanged: the mean number of post-90 goals per match in 2023–24 is significantly higher than across the previous two seasons. The Welch test therefore reproduces the original conclusion using a more appropriate inferential method.

Summary statistics for cards across 21/22 and 22/23

```

cards_21_23 <- summary_by_match %>%
  filter(Season %in% c("2021-22", "2022-23"))

mean(cards_21_23$card_count)

```

```

## [1] 3.657895

sd(cards_21_23$card_count)

## [1] 2.030418

t.test(
  cards_21_23$card_count,
  conf.level = 0.95
)

##
## One Sample t-test
##
## data: cards_21_23$card_count
## t = 49.665, df = 759, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 3.513311 3.802479
## sample estimates:
## mean of x
## 3.657895

cards_23_24 <- summary_by_match %>%
  filter(Season == "2023-24")

mean(cards_23_24$card_count)

## [1] 4.444737

sd(cards_23_24$card_count)

## [1] 2.287882

subs_23_24 <- summary_by_match %>%
  filter(Season == "2023-24")

mean(subs_23_24$substitution_count)

## [1] 7.960526

sd(subs_23_24$substitution_count)

## [1] 1.571857

subs_22_23 <- summary_by_match %>%
  filter(Season == "2022-23")

mean(subs_22_23$substitution_count)

## [1] 7.855263

sd(subs_22_23$substitution_count)

```

```
## [1] 1.522946
```

A 95% confidence interval is calculated for the 2022–23 substitution rate.

```
t.test(
  subs_22_23$substitution_count,
  conf.level = 0.95
)

##
## One Sample t-test
##
## data: subs_22_23$substitution_count
## t = 100.55, df = 379, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 7.701649 8.008877
## sample estimates:
## mean of x
## 7.855263
```

Mean and standard deviation of post-90 substitutions for 22/23 and 23/24

```
subs_90_23_24 <- summary_count_90_complete %>%
  filter(Season == "2023-24")

mean(subs_90_23_24$substitution_count)

## [1] 0.5289474

sd(subs_90_23_24$substitution_count)

## [1] 0.8197457

subs_90_22_23 <- summary_count_90_complete %>%
  filter(Season == "2022-23")

mean(subs_90_22_23$substitution_count)

## [1] 0.4078947

sd(subs_90_22_23$substitution_count)

## [1] 0.7404205

t.test(
  subs_90_22_23$substitution_count,
  conf.level = 0.95
)

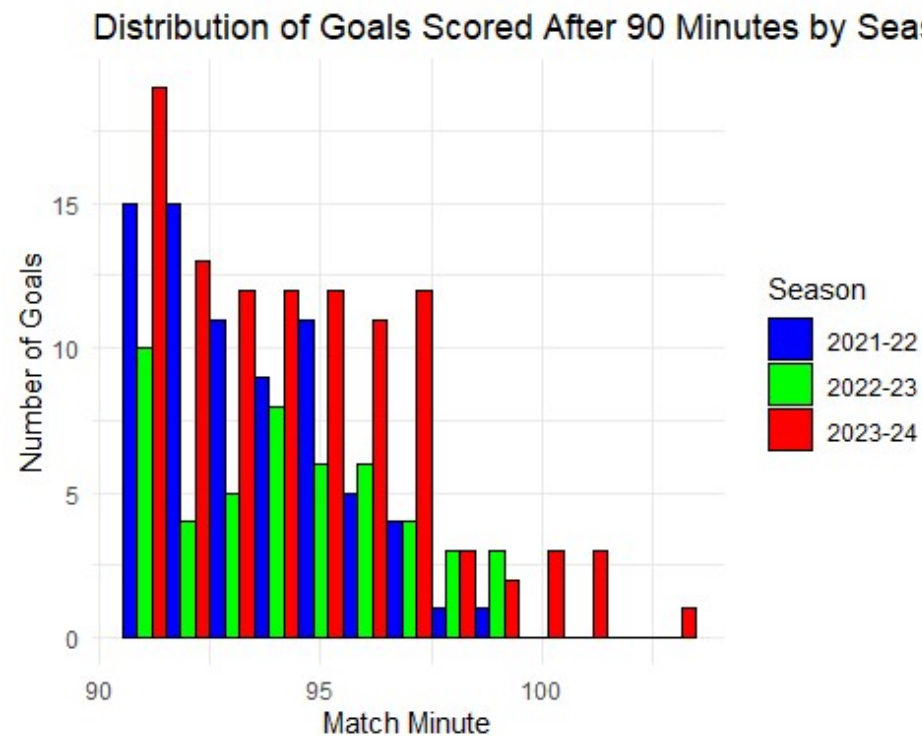
##
## One Sample t-test
##
```

```
## data: subs_90_22_23$substitution_count
## t = 10.739, df = 379, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 0.3332114 0.4825781
## sample estimates:
## mean of x
## 0.4078947
```

To compare the timing of late goals across all three seasons, the post-90 goal distributions are plotted together and separately by season.

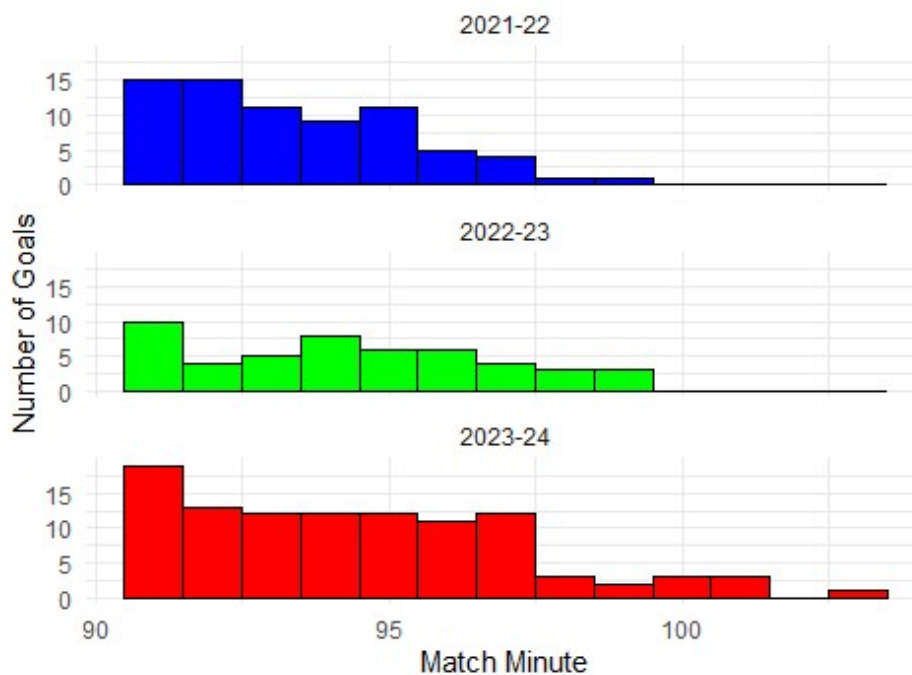
```
goals_after_90_all <- combined_matchevents %>%
  filter(
    Event_Minute > 90,
    Event_Type %in% c(
      "Goal",
      "Own Goal",
      "label.penalty.scored"
    )
  )

ggplot(
  goals_after_90_all,
  aes(x = Event_Minute, fill = Season)
) +
  geom_histogram(
    binwidth = 1,
    position = "dodge",
    color = "black"
  ) +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Distribution of Goals Scored After 90 Minutes by Season",
    x = "Match Minute",
    y = "Number of Goals",
    fill = "Season"
  ) +
  theme_minimal()
```



```
ggplot(
  goals_after_90_all,
  aes(x = Event_Minute, fill = Season)
) +
  geom_histogram(
    binwidth = 1,
    color = "black",
    show.legend = FALSE
  ) +
  facet_wrap(~ Season, ncol = 1) +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Distribution of Goals Scored After 90 Minutes by Season",
    x = "Match Minute",
    y = "Number of Goals"
  ) +
  theme_minimal()
```

Distribution of Goals Scored After 90 Minutes by Season



```
# Count goals scored after 90 minutes by season
goals_after_90_summary <- combined_matchevents %>%
  filter(
    Event_Minute > 90,
    Event_Type %in% c("Goal", "Own Goal", "label.penalty.scored")
  ) %>%
  count(Season, name = "goals_after_90") %>%
  rename(season = Season)

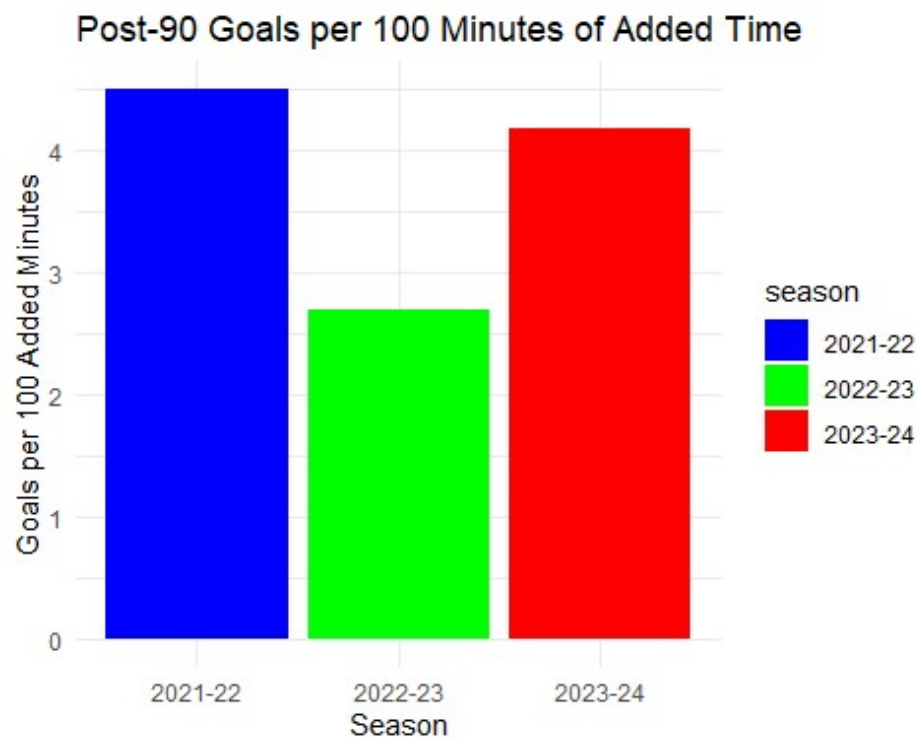
# Calculate goals per 100 minutes of second-half added time
late_goal_rate <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    total_added_minutes = sum(`90+`),
    .groups = "drop"
  ) %>%
  left_join(
    goals_after_90_summary,
    by = "season"
  ) %>%
  mutate(
    goals_per_added_minute =
      goals_after_90 / total_added_minutes,
    goals_per_100_added_minutes =
      100 * goals_per_added_minute
  )
```

```
late_goal_rate
```

```
## # A tibble: 3 × 5
```

```
##   season total_added_minutes goals_after_90 goals_per_added_minute
##   <chr>         <dbl>         <int>         <dbl>
## 1 2021-22         1603             72         0.0449
## 2 2022-23         1823             49         0.0269
## 3 2023-24         2465            103         0.0418
## # i 1 more variable: goals_per_100_added_minutes <dbl>
```

```
ggplot(
  late_goal_rate,
  aes(
    x = season,
    y = goals_per_100_added_minutes,
    fill = season
  )
) +
  geom_col() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Post-90 Goals per 100 Minutes of Added Time",
    x = "Season",
    y = "Goals per 100 Added Minutes"
  ) +
  theme_minimal()
```



To relate the scoring changes to betting markets, I first calculate the proportion of matches finishing with more than 2.5 total goals in each season.

```
over_25_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    matches = n(),
    over_25 = sum(total_goals > 2.5),
    over_25_percentage = 100 * over_25 / matches
  )

over_25_summary

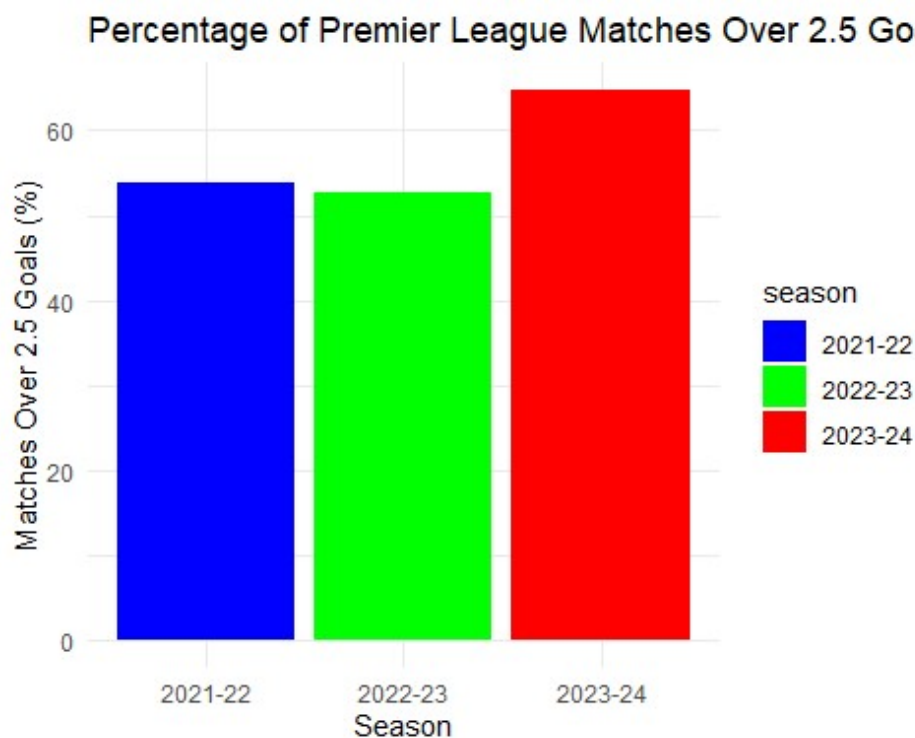
## # A tibble: 3 × 4
##   season matches over_25 over_25_percentage
##   <chr>   <int>   <int>         <dbl>
## 1 2021-22   380    205         53.9
## 2 2022-23   380    200         52.6
## 3 2023-24   380    246         64.7

ggplot(
  over_25_summary,
  aes(x = season, y = over_25_percentage, fill = season)
) +
  geom_col() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  ))
```

```

  "2022-23" = "green",
  "2023-24" = "red"
)) +
labs(
  title = "Percentage of Premier League Matches Over 2.5 Goals",
  x = "Season",
  y = "Matches Over 2.5 Goals (%)"
) +
theme_minimal()

```



I also calculate the proportion of matches in which both teams scored (BTTS).

```

btts_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    total_matches = n(),
    btts_matches = sum(
      home_team_goals > 0 &
      away_team_goals > 0
    ),
    btts_percentage = 100 * btts_matches / total_matches
  )

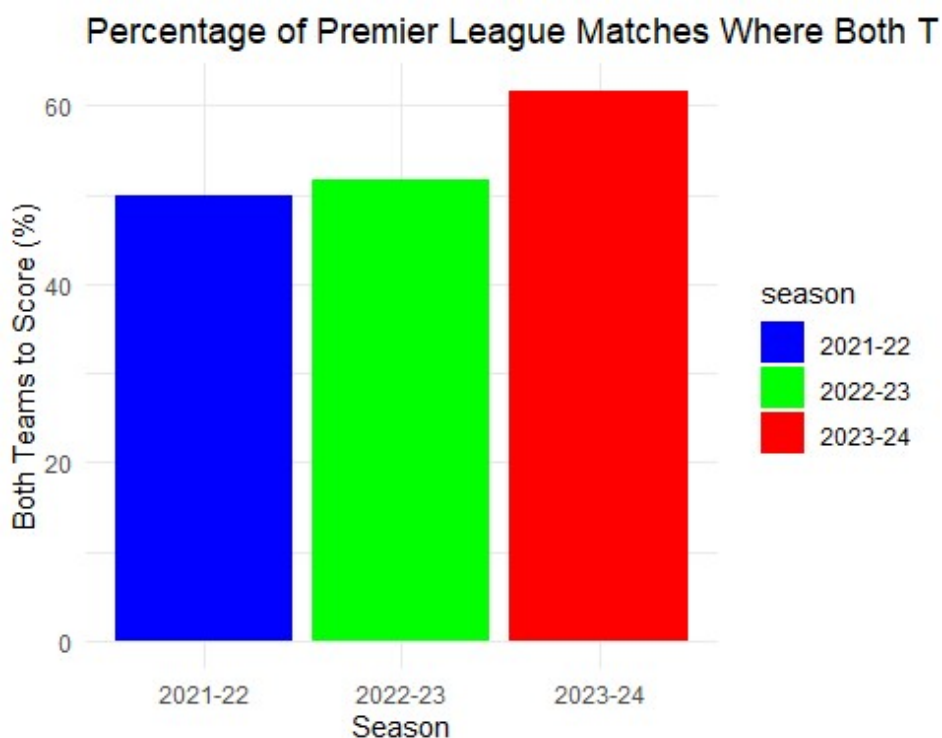
btts_summary

## # A tibble: 3 × 4
##   season total_matches btts_matches btts_percentage

```

```
##   <chr>           <int>      <int>      <dbl>
## 1 2021-22         380        190        50
## 2 2022-23         380        196       51.6
## 3 2023-24         380        234       61.6

ggplot(
  btts_summary,
  aes(x = season, y = btts_percentage, fill = season)
) +
  geom_col() +
  scale_fill_manual(values = c(
    "2021-22" = "blue",
    "2022-23" = "green",
    "2023-24" = "red"
  )) +
  labs(
    title = "Percentage of Premier League Matches Where Both Teams Scored",
    x = "Season",
    y = "Both Teams to Score (%)"
  ) +
  theme_minimal()
```



The proportion of matches in which both teams scored also increased substantially in 2023–24. This suggests that the rise in total scoring was not limited to a small number of one-sided, high-scoring matches. From a betting perspective, the increase in BTTS frequency is relevant when estimating probabilities for this market in later seasons.

```

betting_summary <- combined_matchdata %>%
  group_by(season) %>%
  summarise(
    Over_2_5 = 100 * mean(total_goals > 2.5),
    BTTS = 100 * mean(
      home_team_goals > 0 &
      away_team_goals > 0
    )
  )

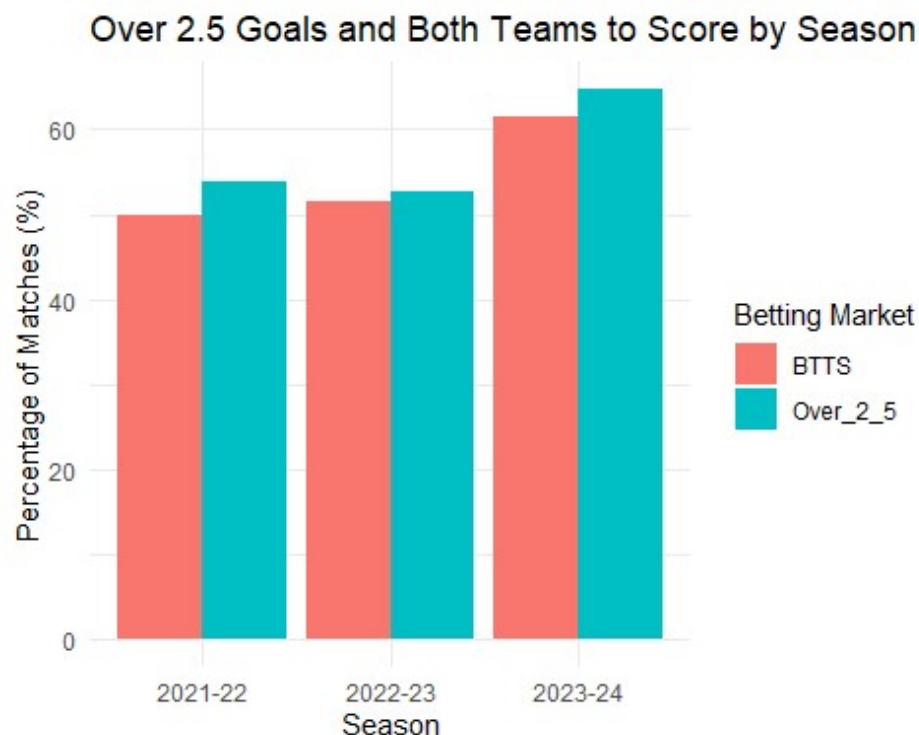
betting_summary

## # A tibble: 3 × 3
##   season Over_2_5 BTTS
##   <chr>   <dbl> <dbl>
## 1 2021-22    53.9    50
## 2 2022-23    52.6    51.6
## 3 2023-24    64.7    61.6

betting_summary_long <- betting_summary %>%
  pivot_longer(
    cols = c(Over_2_5, BTTS),
    names_to = "Market",
    values_to = "Percentage"
  )

ggplot(
  betting_summary_long,
  aes(x = season, y = Percentage, fill = Market)
) +
  geom_col(position = "dodge") +
  labs(
    title = "Over 2.5 Goals and Both Teams to Score by Season",
    x = "Season",
    y = "Percentage of Matches (%)",
    fill = "Betting Market"
  ) +
  theme_minimal()

```



```
betting_summary_long <- betting_summary_long %>%
  mutate(
    Market = recode(
      Market,
      "Over_2_5" = "Over 2.5 Goals",
      "BTTTS" = "Both Teams to Score"
    )
  )
```

The longer match durations observed in 2023–24 appear to have contributed to the rise in scoring, post-90 goals became substantially more common, both in total and across individual matches. However, longer added time cannot fully explain the higher-scoring season: only around one-third of the increase in goals from 2022–23 to 2023–24 occurred after the 90th minute. The simultaneous rise in Over 2.5 and BTTTS outcomes therefore suggests a broader change in the scoring environment, rather than an effect explained solely by additional stoppage time.

From a betting perspective, such a change would be relevant to markets including Over 2.5 Goals and Both Teams to Score. Predictive models or bookmaker probabilities based primarily on older seasons may underestimate these outcomes if changes in the underlying scoring environment persist. However, with only three seasons of event-level data, the analysis identifies an important shift rather than establishing that the change will continue into future seasons.

Extension to 24/25 and 25/26 with consistent datasets

```

raw_2021_22 <- read_csv("2122.csv")
raw_2022_23 <- read_csv("2223.csv")
raw_2023_24 <- read_csv("2324.csv")
raw_2024_25 <- read_csv("2425.csv")
raw_2025_26 <- read_csv("2526.csv")

```

The Football-Data match files are used to extract match results and Bet365 Over/Under 2.5 goal odds consistently across all five seasons.

```

odds2024_25 <- raw_2024_25 %>%
  transmute(
    season = "2024-25",
    home_team = HomeTeam,
    away_team = AwayTeam,
    home_team_goals = FTHG,
    away_team_goals = FTAG,
    total_goals = FTHG + FTAG,
    over_25_odds = `B365>2.5`,
    under_25_odds = `B365<2.5`
  )

```

```

odds2025_26 <- raw_2025_26 %>%
  transmute(
    season = "2025-26",
    home_team = HomeTeam,
    away_team = AwayTeam,
    home_team_goals = FTHG,
    away_team_goals = FTAG,
    total_goals = FTHG + FTAG,
    over_25_odds = `B365>2.5`,
    under_25_odds = `B365<2.5`
  )

```

```

odds2021_22 <- raw_2021_22 %>%
  transmute(
    season = "2021-22",
    home_team = HomeTeam,
    away_team = AwayTeam,
    home_team_goals = FTHG,
    away_team_goals = FTAG,
    total_goals = FTHG + FTAG,
    over_25_odds = `B365>2.5`,
    under_25_odds = `B365<2.5`
  )

```

```

odds2022_23 <- raw_2022_23 %>%
  transmute(
    season = "2022-23",
    home_team = HomeTeam,
    away_team = AwayTeam,

```

```

    home_team_goals = FTHG,
    away_team_goals = FTAG,
    total_goals = FTHG + FTAG,
    over_25_odds = `B365>2.5`,
    under_25_odds = `B365<2.5`
  )

odds2023_24 <- raw_2023_24 %>%
  transmute(
    season = "2023-24",
    home_team = HomeTeam,
    away_team = AwayTeam,
    home_team_goals = FTHG,
    away_team_goals = FTAG,
    total_goals = FTHG + FTAG,
    over_25_odds = `B365>2.5`,
    under_25_odds = `B365<2.5`
  )

all_match_odds <- bind_rows(
  odds2021_22,
  odds2022_23,
  odds2023_24,
  odds2024_25,
  odds2025_26)

dim(all_match_odds)

## [1] 1900      8

table(all_match_odds$season)

##
## 2021-22 2022-23 2023-24 2024-25 2025-26
##      380      380      380      380      380

all_match_odds <- all_match_odds %>%
  mutate(
    over_25 = total_goals > 2.5,
    btts = home_team_goals > 0 & away_team_goals > 0,
    implied_over = 1 / over_25_odds,
    implied_under = 1 / under_25_odds,
    market_prob_over_25 =
      implied_over / (implied_over + implied_under)
  )

season_odds_summary <- all_match_odds %>%
  group_by(season) %>%
  summarise(
    mean_goals = mean(total_goals),
    over_25_percentage = 100 * mean(over_25),

```

```

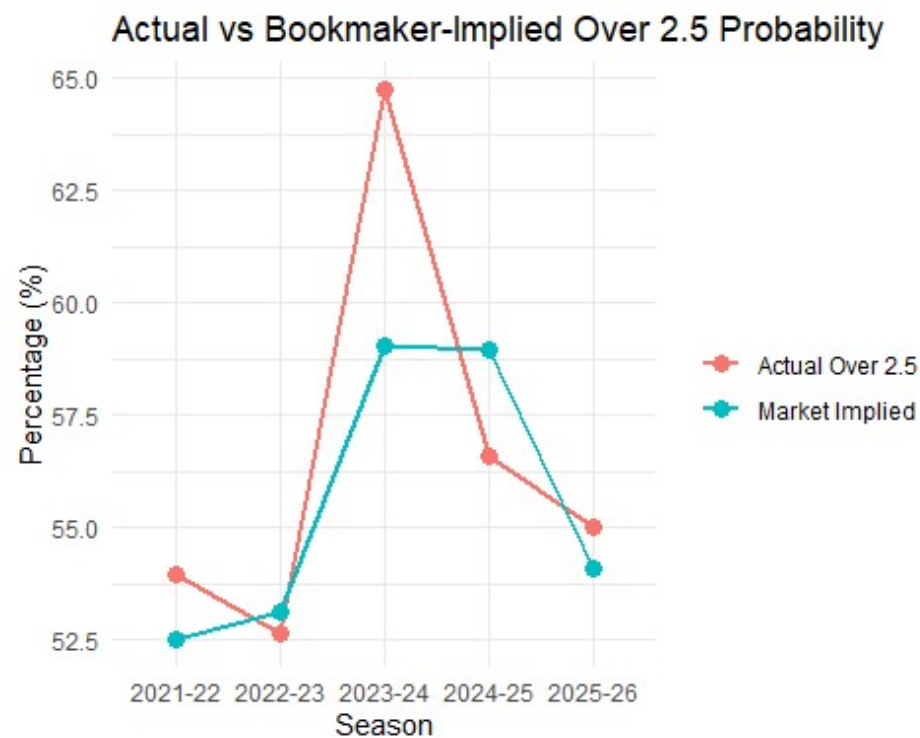
    bttb_percentage = 100 * mean(bttb),
    market_over_25_percentage =
      100 * mean(market_prob_over_25, na.rm = TRUE)
  )

season_odds_summary

## # A tibble: 5 × 5
##   season mean_goals over_25_percentage bttb_percentage market_over_25_per
cent...1
##   <chr>      <dbl>          <dbl>          <dbl>
<dbl>
## 1 2021-22      2.82          53.9          50
52.5
## 2 2022-23      2.85          52.6          51.6
53.1
## 3 2023-24      3.28          64.7          61.6
59.1
## 4 2024-25      2.93          56.6          57.4
59.0
## 5 2025-26      2.75          55          56.1
54.1
## # i abbreviated name: 1market_over_25_percentage

ggplot(season_odds_summary, aes(x = season, group = 1)) +
  geom_line(
    aes(y = over_25_percentage, colour = "Actual Over 2.5"),
    linewidth = 1
  ) +
  geom_point(
    aes(y = over_25_percentage, colour = "Actual Over 2.5"),
    size = 3
  ) +
  geom_line(
    aes(y = market_over_25_percentage, colour = "Market Implied"),
    linewidth = 1
  ) +
  geom_point(
    aes(y = market_over_25_percentage, colour = "Market Implied"),
    size = 3
  ) +
  labs(
    title = "Actual vs Bookmaker-Implied Over 2.5 Probability",
    x = "Season",
    y = "Percentage (%)",
    colour = ""
  ) +
  theme_minimal()

```



```
season_odds_summary <- season_odds_summary %>%
  mutate(
    market_error =
      over_25_percentage - market_over_25_percentage
  )

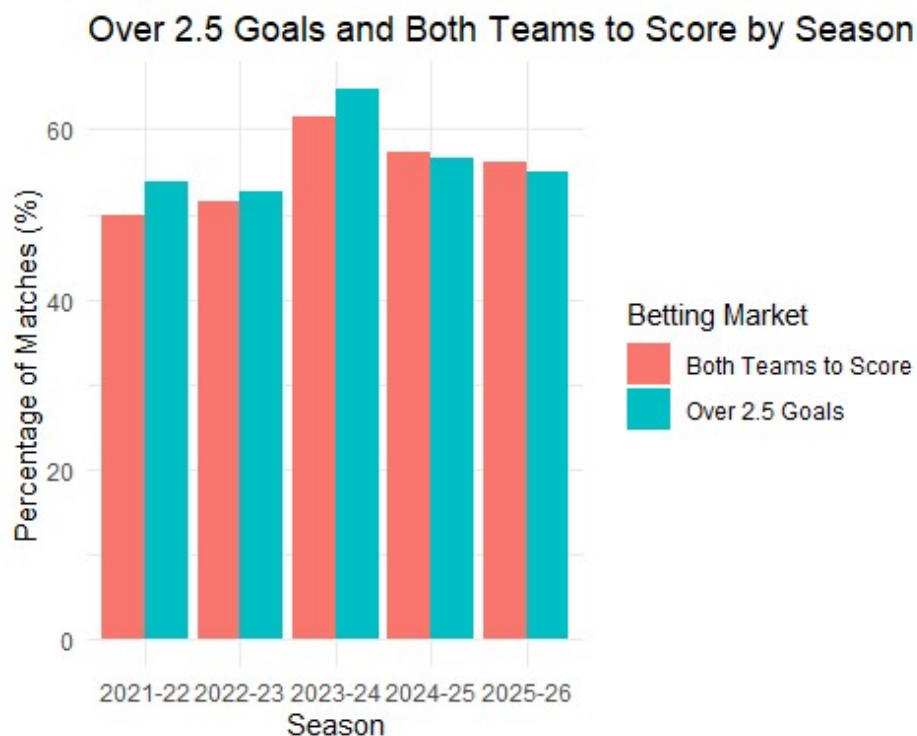
season_odds_summary

## # A tibble: 5 × 6
##   season mean_goals over_25_percentage btts_percentage market_over_25_per
cent...1
##   <chr>      <dbl>          <dbl>          <dbl>
<dbl>
## 1 2021-22      2.82          53.9           50
52.5
## 2 2022-23      2.85          52.6           51.6
53.1
## 3 2023-24      3.28          64.7           61.6
59.1
## 4 2024-25      2.93          56.6           57.4
59.0
## 5 2025-26      2.75           55           56.1
54.1
## # i abbreviated name: 1market_over_25_percentage
## # i 1 more variable: market_error <dbl>
```

```

betting_summary_long <- season_odds_summary %>%
  select(season, over_25_percentage, btts_percentage) %>%
  pivot_longer(
    cols = c(over_25_percentage, btts_percentage),
    names_to = "Market",
    values_to = "Percentage"
  ) %>%
  mutate(
    Market = recode(
      Market,
      "over_25_percentage" = "Over 2.5 Goals",
      "btts_percentage" = "Both Teams to Score"
    )
  )
ggplot(
  betting_summary_long,
  aes(x = season, y = Percentage, fill = Market)
) +
  geom_col(position = "dodge") +
  labs(
    title = "Over 2.5 Goals and Both Teams to Score by Season",
    x = "Season",
    y = "Percentage of Matches (%)",
    fill = "Betting Market"
  ) +
  theme_minimal()

```



Extending the analysis to 2024–25 and 2025–26 using the Football-Data match files showed that both Over 2.5 and BTTS outcomes peaked in 2023–24 before falling back in the next two seasons. This suggests that 2023–24 was an unusually high-scoring season rather than clear evidence of a permanently shifted scoring environment.

This supports the idea that longer match durations in 2023–24 were related to the increase in scoring, but they do not appear to explain a lasting structural change across all later seasons.

The Premier League supply their own match length data: 2021–22: 97:29 2022–23: 98:27 2023–24: 101:36 2024–25: 99:43 2025–26: 100:20

ref: [https://www.premierleague.com/en/news/4271974?](https://www.premierleague.com/en/news/4271974?https://www.premierleague.com/en/news/4580269/why-more-late-goals-are-being-scored-in-2025-26-premier-league-season?)
<https://www.premierleague.com/en/news/4580269/why-more-late-goals-are-being-scored-in-2025-26-premier-league-season?>

This tells us that matches didn't keep getting longer, 23/24 was unusually long, which is consistent with the goals scored stats pattern we have found (over 2.5 goals being highest in this season as well). This is consistent with the hypothesis that longer matches explain general rise in scoring, but this is still only an association, and in 25/26 we see match length increase from the drop in the previous season, but a drop in percentage of matches with over 2.5 goals. We can say that there is some evidence in favour of the original hypothesis being investigated, longer matches contributed to the exceptional number of late goals in 23/24. Unfortunately due to the event data the original author used not being readily available for 24/25 and 25/26 seasons, we could not extend the analyses further.

It is worth stating that the datasets used were suitable for the investigation into correlation between post90 goals and increased stoppage time, although it is unclear where the datasets came from, and extensions into later seasons would have been helpful.

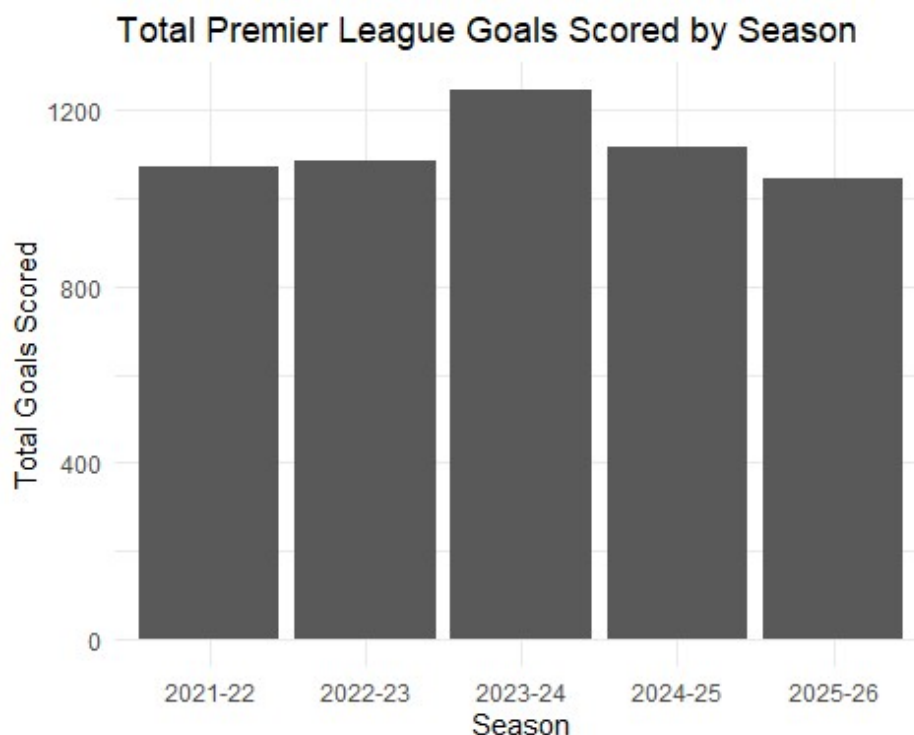
extend analysis of total goals scored

```
total_goals_summary <- all_match_odds %>%
  group_by(season) %>%
  summarise(
    total_goals_scored = sum(total_goals),
    mean_goals_per_match = mean(total_goals)
  )

total_goals_summary

## # A tibble: 5 × 3
##   season total_goals_scored mean_goals_per_match
##   <chr>         <dbl>         <dbl>
## 1 2021-22         1071           2.82
## 2 2022-23         1084           2.85
## 3 2023-24         1246           3.28
## 4 2024-25         1115           2.93
## 5 2025-26         1045           2.75
```

```
ggplot(
  total_goals_summary,
  aes(x = season, y = total_goals_scored)
) +
  geom_col() +
  labs(
    title = "Total Premier League Goals Scored by Season",
    x = "Season",
    y = "Total Goals Scored"
  ) +
  theme_minimal()
```



```
season_summary <- all_match_odds %>%
  group_by(season) %>%
  summarise(
    total_goals = sum(total_goals),
    mean_goals = mean(total_goals),
    over_25_percentage = 100 * mean(over_25),
    btts_percentage = 100 * mean(btts),
    market_over_25_percentage =
      100 * mean(market_prob_over_25, na.rm = TRUE)
  )

season_summary

## # A tibble: 5 × 6
##   season total_goals mean_goals over_25_percentage btts_percentage
```

```
##      <chr>          <dbl>      <dbl>          <dbl>          <dbl>
## 1 2021-22          1071        1071          53.9           50
## 2 2022-23          1084        1084          52.6           51.6
## 3 2023-24          1246        1246          64.7           61.6
## 4 2024-25          1115        1115          56.6           57.4
## 5 2025-26          1045        1045           55           56.1
## # i 1 more variable: market_over_25_percentage <dbl>
```

```
season_summary <- all_match_odds %>%
  group_by(season) %>%
  summarise(
    total_goals = sum(total_goals),
    mean_goals = round(mean(total_goals), 2),
    over_25_percentage = round(100 * mean(over_25), 1),
    btts_percentage = round(100 * mean(btts), 1),
    market_over_25_percentage =
      round(100 * mean(market_prob_over_25, na.rm = TRUE), 1)
  )
```

```
season_summary
```

```
## # A tibble: 5 × 6
##   season total_goals mean_goals over_25_percentage btts_percentage
##   <chr>      <dbl>      <dbl>          <dbl>          <dbl>
## 1 2021-22    1071        1071          53.9           50
## 2 2022-23    1084        1084          52.6           51.6
## 3 2023-24    1246        1246          64.7           61.6
## 4 2024-25    1115        1115          56.6           57.4
## 5 2025-26    1045        1045           55           56.1
## # i 1 more variable: market_over_25_percentage <dbl>
```